Graphic Era

# Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no. 2594102

Name of the Course: **B. Tech (AIML and AIDS)**
Semester: II
Name of the Paper: Fundamentals of Artificial Intelligence and Machine Learning
Paper Code: TCS203
**Time: 1.5 hour**
Maximum Marks: 50

**Note:**
(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Define PEAS description and explain its components with an example (e.g., autonomous ca1r or vacuum cleaner). **CO1**

OR

b. Discuss the impact of Artificial Intelligence on society and industry. Discuss both positive and negative implications. **CO1**

Q2. (10 Marks)

a. Explain the working of Depth-First Search (DFS) algorithm with the help of an example and analyse the algorithm based on the following: **CO2**
   i. Completeness
   ii. Optimality
   iii. Time Complexity

OR

b. Consider the following state space graph consisting of five nodes:
   Nodes: S (Start), A, B, C, G (Goal) **CO2**
   Edges:
   S → A
   S → B
   A → C
   B → E
   E → D
   D → G

   i. Apply the Breadth-First Search (BFS) algorithm to find a path from S to G.
   ii. Identify the final path obtained.
   iii. Comment on whether BFS guarantees the optimal solution in this case.

Q3. (10 Marks)

a. Discuss Intelligent Agents and their types and describe how they interact with environment. **CO1**

OR

b. Consider the following weighted graph with six nodes node A as start and node F as goal: **CO2**
Nodes: **A, B, C, D, E, F**
Edge Costs:

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- A → B = 2
- A → C = 3
- B → D = 4
- C → D = 1
- C → E = 5
- D → F = 2
- E → F = 2

| Node | h(n) |
|---|---|
| A | 6 |
| B | 5 |
| C | 3 |
| D | 2 |
| E | 2 |
| F | 0 |

i. Apply the A* algorithm to find the optimal path from A to F.
ii. Compute the values of g(n), h(n), and f(n)=g(n)+h(n) at each step.
iii. Identify the final optimal path and total path cost.

Q4. (10 Marks)

a. i. Discuss what data cleaning is and the need of cleaning the data before applying machine learning.
ii. Mention any 3 different types of impurities present in the dataset and how to remove them. **CO3**

OR

b. Explain Agent Architecture in detail. Discuss different types of environments with real-life examples. **CO1**

Q5. (10 Marks)

a. A retail company wants to predict customer purchase behaviour using the following dataset and perform the following: **CO3**
i. Identify and handle the missing values.
ii. Detect and remove outliers in monthly spend using the z-score technique.
iii. After cleaning the workout hours column, assume workout hours = [5, 6, 4, 5], apply min-max normalization on workout hours column.
iv. Given standard deviation as 1.48, calculate the standardized value for age = 19 by applying standard scalar after removing inconsistency in age column.

| Customer ID | Age | Monthly Spend | Workout Hours | Membership Type |
|---|---|---|---|---|
| 1 | 21 | 1500 | 5 | Basic |
| 2 | -5 | 1800 | 6 | Premium |
| 3 | 20 | 12000 | 4 | Basic |
| 4 | 18 | 1600 | NULL | Premium |
| 5 | 26 | 1550 | 5 | Basic |

OR

b. Explain the difference between the following with the help of examples: **CO3**
i. AI, ML, Deep Learning
ii. Supervised and Unsupervised Learning